

Portfolio Case Study – Fedor Gorbunov

Real-Time FastAPI Backend for Market Data & AI Analytics

Executive Summary

I designed and built a private real-time market data backend from architecture to production operation. The system processes minute-level market data, maintains calculated analytics, validates results, stores ML-ready history, and protects downstream analysis from corrupted or untrusted source data.

Business Problem

A live analytical system needs more than raw data ingestion. It needs continuous time-series integrity, fast recovery after restarts, duplicate/gap prevention, calculated features, history coverage, and a reliable way to know whether stored data can be trusted before downstream processing continues.

What I Built

The pipeline ingests live Binance market data, uses Redis as a runtime buffer, stores base and calculated rows in MongoDB, maintains ML-ready mirrors, exports parquet snapshots, and provides machine-generated reports. Health monitoring tracks service readiness, data freshness, storage coverage, and runtime status.

Architecture Highlights

- Binance WS/REST → Redis runtime buffers → MongoDB base/calculated collections.
- ML-ready mirror and parquet snapshots for offline analysis.
- Health monitor for Binance, Redis, MongoDB, REST, WebSocket, ZeroMQ, and calculation services.
- Independent validation layer for calculated rows and derived features.
- History coverage matrix for retention, covered days, missing days, row counts, and storage proof.

Reliability Model

Binance is used as an external trust source. If stored base candles fail the oracle check, downstream processing is blocked and recovery/recalculation starts. Startup gates check Redis, MongoDB, storage readiness, and runtime prerequisites before accepting the system as healthy.

My Role

Systems Architect, Backend Developer, Data Pipeline Operator. I designed the data lifecycle, recovery model, validation workflow, monitoring surface, and production operation logic.

Result

A continuous minute-level data pipeline with source-trust guards, anti-duplicate safeguards, recovery flows, ML-ready history, validation views, and machine-generated reports. The value is reliable data infrastructure, not trading-income claims.

Not Disclosed

Trading rules, strategy thresholds, proprietary formulas, private calculations, API credentials, account data, and sensitive strategy details.